

WOC 8.0

Project Report by AniketKumar

Linear Regression

Formulas used:

Cost:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Gradient:

$$\frac{\partial J}{\partial w} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

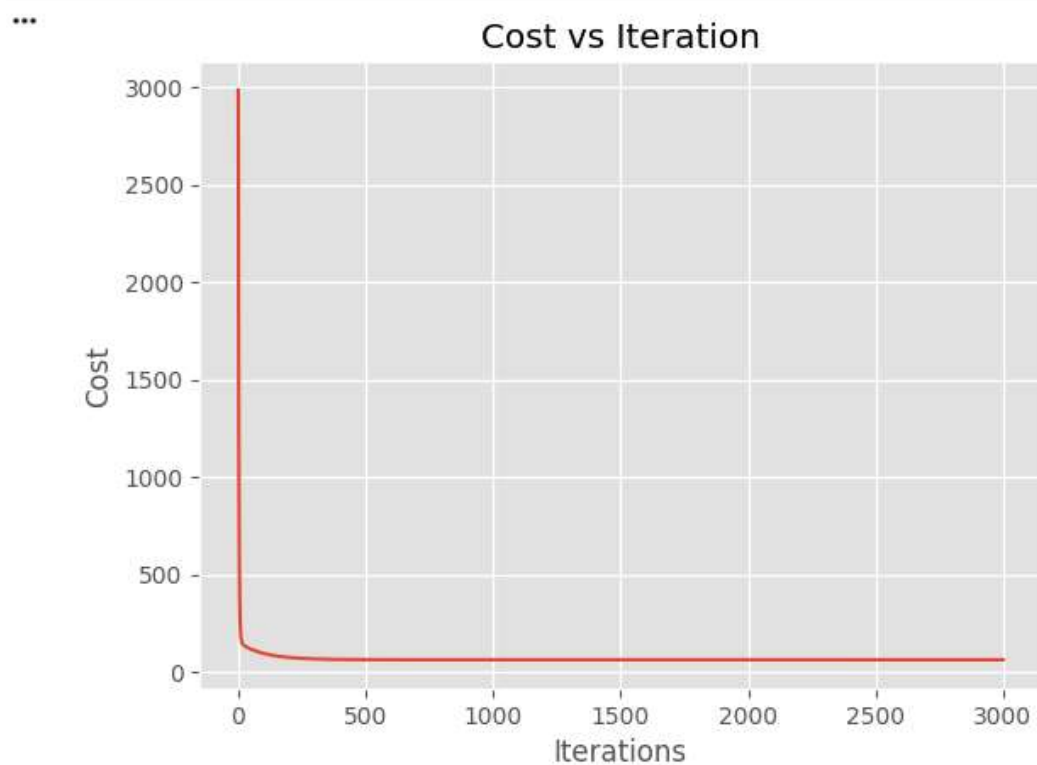
$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

Graphs:

```
alpha=0.2
iteration=3000
wf,bf,j = gradient_descent(X_train_norm,y_train,w,b,alpha,iteration)
```

```
Iteration 0: cost 2986.2272056686697
Iteration 300: cost 65.74288607420192
Iteration 600: cost 61.76291827981492
Iteration 900: cost 61.574579626405864
Iteration1200: cost 61.56561801766702
Iteration1500: cost 61.565190977693725
Iteration1800: cost 61.56517062040965
Iteration2100: cost 61.565169649864224
Iteration2400: cost 61.56516960359164
Iteration2700: cost 61.56516960138549
```

▶ cost_vs_iteration(j)



Info:

R2 score on training data: 0.9683

R2 score on cross-validation data: 0.9690

- ☐ I have used vectorization.
- ☐ I also normalized the data

Polynomial Regression

Formulas used:

Cost:

$$J(\vec{w}, b) = \frac{1}{2m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

Gradient:

$$\frac{\partial J(\mathbf{w}, b)}{\partial w_j} = \frac{1}{m} \sum_{i=0}^{m-1} (f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{m} w_j$$

$$\frac{\partial J(\mathbf{w}, b)}{\partial b} = \frac{1}{m} \sum_{i=0}^{m-1} (f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)})$$

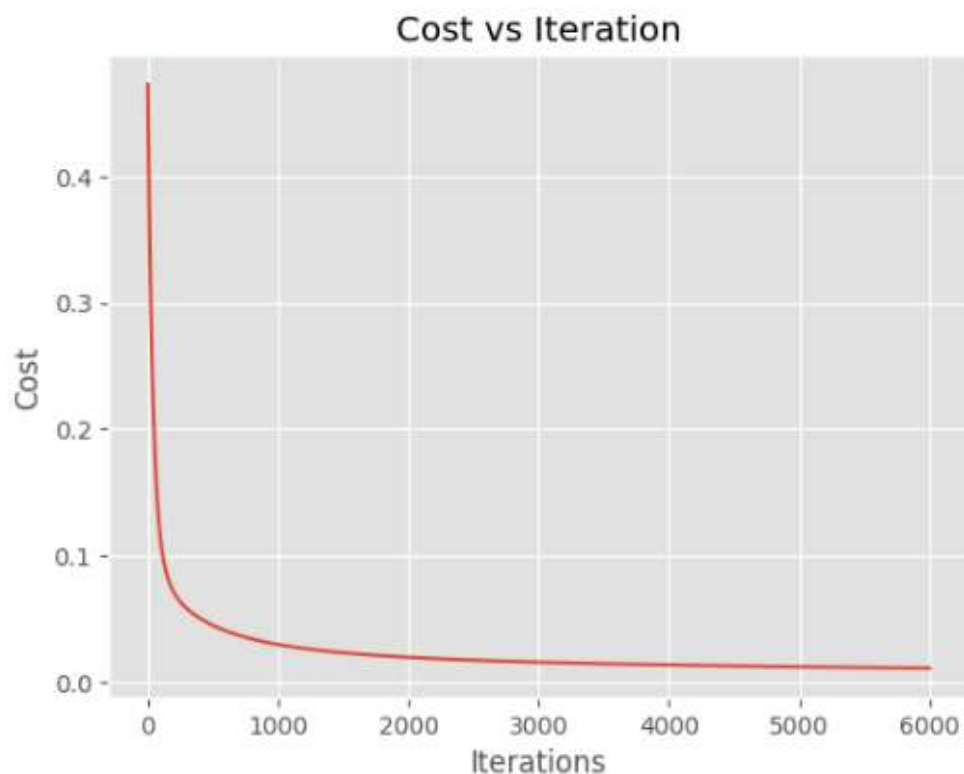
Graphs:

```
alpha=0.001
iteration=6000
lambda_=0.1
wf,bf,j=gradient_descent(X_train_data,Y_train_data,w,b,alpha,iteration, lambda_)
```

```
Iteration 0: Cost 0.47309272953002873
Iteration 600: Cost 0.04035560807903367
Iteration 1200: Cost 0.02646218149857529
Iteration 1800: Cost 0.02072611999631158
Iteration 2400: Cost 0.017605034004364934
Iteration 3000: Cost 0.015608513566896237
Iteration 3600: Cost 0.014188119488271825
Iteration 4200: Cost 0.013103042414218873
Iteration 4800: Cost 0.012233798700350414
Iteration 5400: Cost 0.011514819820959182
Iteration 5999: Cost 0.010907757293732225
```

▶ cost_vs_iteration(j)

...



Info:

R2 Score on Training Data: 0.9774692944545276

R2 Score on Cross-Validation Data: 0.9580899986314837

- ☐ I have used degree 4.
- ☐ I have added every combination of features .
- ☐ For numerical stability in gradient I have normalized target values.Because cost where coming in power of $1e40$
- ☐ I used regularization to prevent overfitting.

Logistic Regression

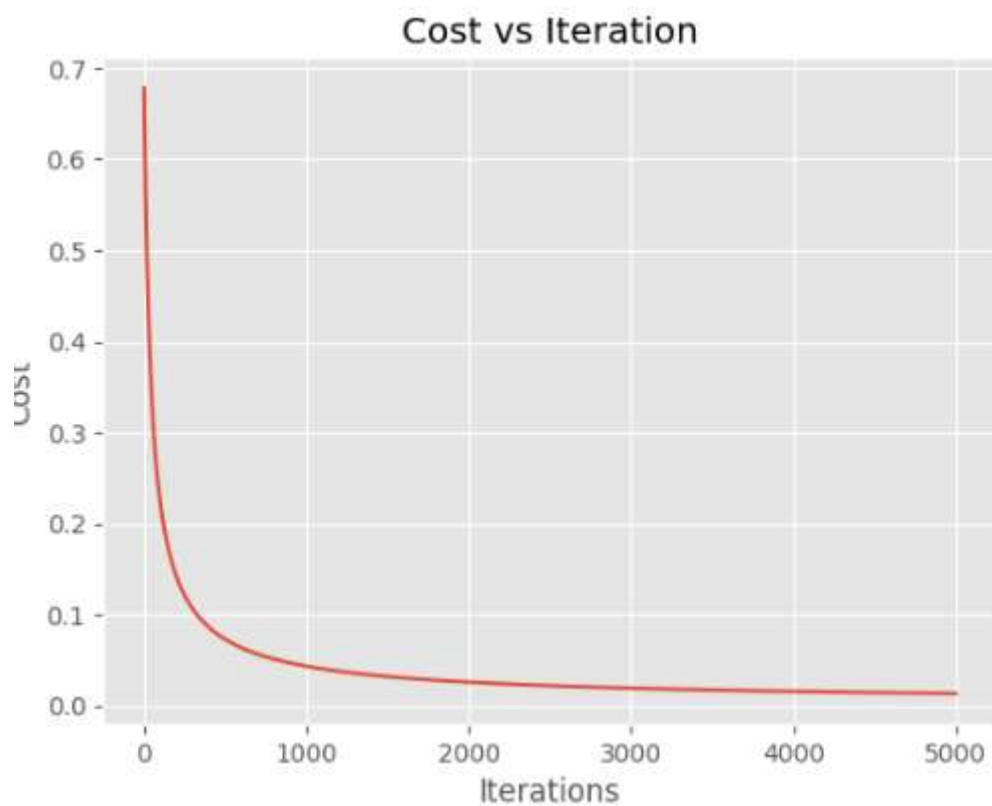
Formulas used:

Same as Linear Regression, just the function f is sigmoid function

Graph:

```
iteration=5000  
alpha=0.02  
w,b,j=gradient_descent(X_train_norm,y_train,w,b,alpha,iteration)
```

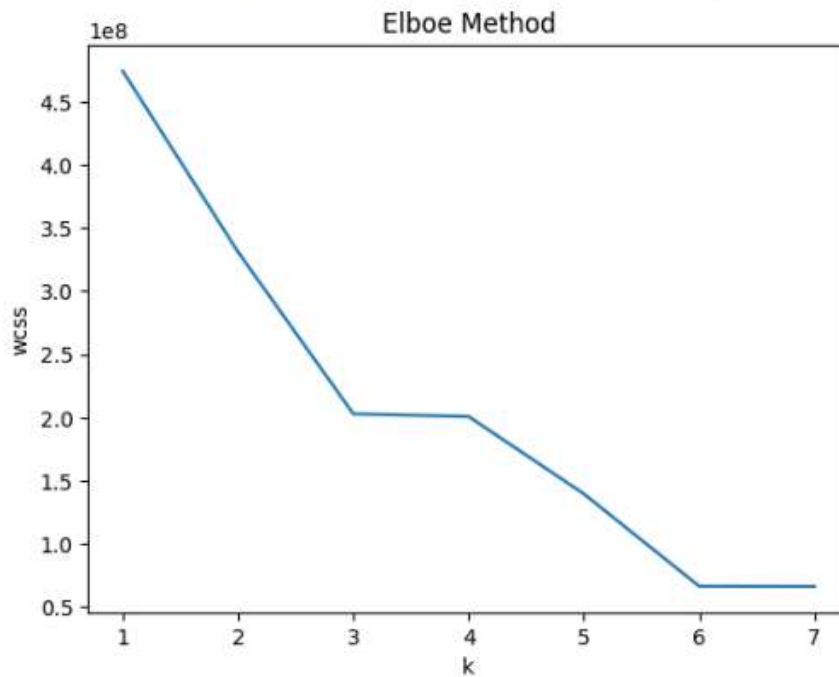
Iteration	0:	Cost	0.6783157866877128
Iteration	500:	Cost	0.072642120781322
Iteration	1000:	Cost	0.043085977058747935
Iteration	1500:	Cost	0.03173980531538217
Iteration	2000:	Cost	0.025596216019852357
Iteration	2500:	Cost	0.021696608827107194
Iteration	3000:	Cost	0.018980499526114896
Iteration	3500:	Cost	0.01696937254999186
Iteration	4000:	Cost	0.015414160424051826
Iteration	4500:	Cost	0.014171842264981504



K-Means

```
elbow(x,7)
```

Number of iterations taken for the cluster centroids to reach the mean centres of 1 clusters: 1
Number of iterations taken for the cluster centroids to reach the mean centres of 2 clusters: 2
Number of iterations taken for the cluster centroids to reach the mean centres of 3 clusters: 20
Number of iterations taken for the cluster centroids to reach the mean centres of 4 clusters: 2
Number of iterations taken for the cluster centroids to reach the mean centres of 5 clusters: 64
Number of iterations taken for the cluster centroids to reach the mean centres of 6 clusters: 16
Number of iterations taken for the cluster centroids to reach the mean centres of 7 clusters: 88



k=3 using the elbow method

N-layer Neural Network

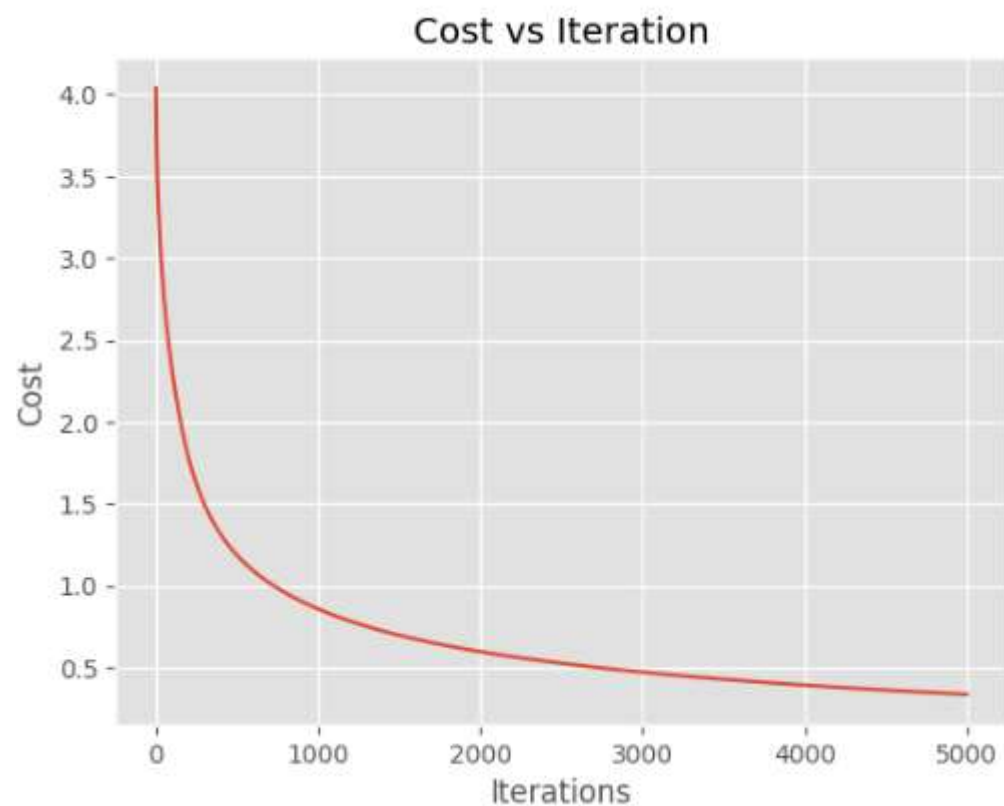
Classification :

Training logs

```
Iter    0 | cost : 4.039145 |  
Iter  500 | cost : 1.186097 |  
Iter 1000 | cost : 0.858271 |  
Iter 1500 | cost : 0.698009 |  
Iter 2000 | cost : 0.597734 |  
Iter 2500 | cost : 0.526964 |  
Iter 3000 | cost : 0.472518 |  
Iter 3500 | cost : 0.429044 |  
Iter 4000 | cost : 0.393285 |  
Iter 4500 | cost : 0.362967 |
```

Macro F1 Score (Train): 0.905732960712467

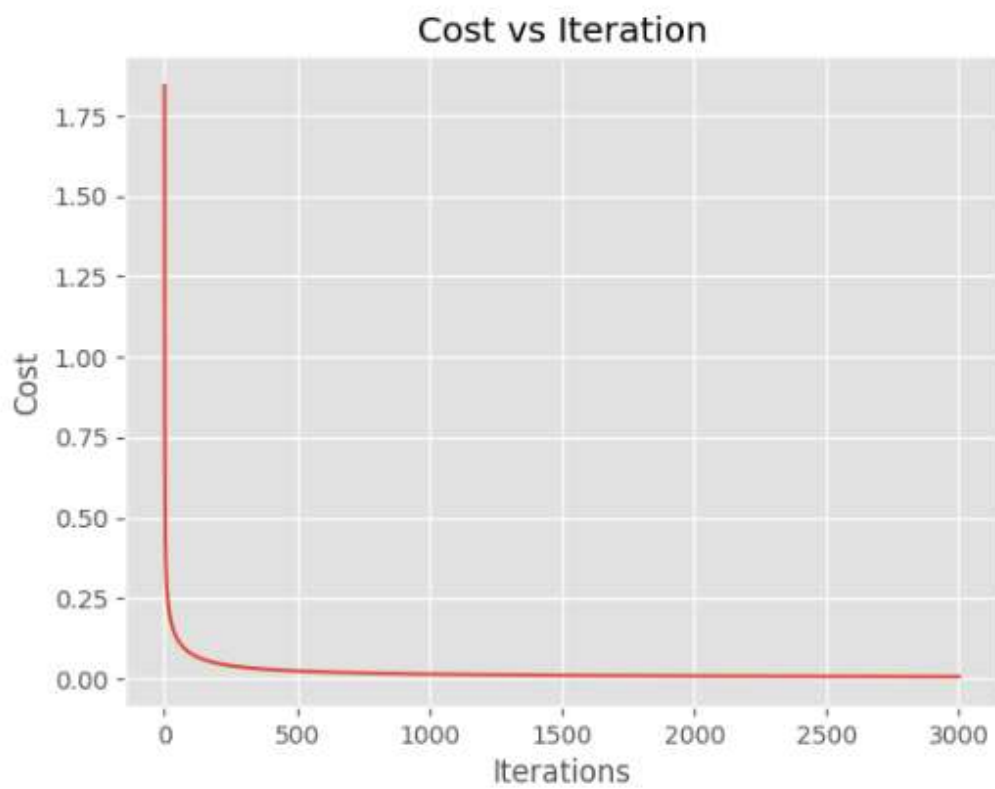
Macro F1 Score(val): 0.8799717986332857



Regression:

Iter	0	cost	: 1.842455
Iter	300	cost	: 0.034756
Iter	600	cost	: 0.020540
Iter	900	cost	: 0.014995
Iter	1200	cost	: 0.011976
Iter	1500	cost	: 0.010024
Iter	1800	cost	: 0.008666
Iter	2100	cost	: 0.007663
Iter	2400	cost	: 0.006887
Iter	2700	cost	: 0.006269

```
cost_vs_iteration(cost_reg)
```



R2 Score (Train): 0.994235327599844

R2 Score (val): 0.9935865779355634

Learning

- Supervised learning (regression,classification)
- Neural Network
- Decision trees
- Optimizers
- Unsupervised learning(clustering)
- EDA
- Why we split data in train ,validation, test
- Vectorization is fast

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